MCK Browser Manager - Google Admin Deployment Guide

This guide walks you through deploying the **MCK Browser Manager** extension to all users in your organization via Google Workspace Admin Console.

Prerequisites

✅ **Google Workspace admin access** with permissions to manage Chrome apps & extensions
✅ **Chrome Web Store Developer account** ($5 one-time registration fee)
✅ **Extension ready to deploy** (this repo contains the complete, tested extension)

Deployment Options

You have **two deployment paths**:

Option A: Chrome Web Store (Recommended)

Upload to Chrome Web Store as **unlisted** (only users with the link can find it) or **private** (only your domain can install it). Google Admin can then force-install it by extension ID.

Pros: Easiest to manage, automatic updates, Google-hosted
Cons: Requires Chrome Web Store developer account, $5 fee, review process (usually 1-3 days)

Option B: Self-Hosted (Advanced)

Host the `.crx` file on your own server and configure Admin Console to install from your URL.

Pros: No Chrome Web Store account needed, full control
Cons: More complex, you manage updates manually, requires HTTPS hosting

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Option A: Chrome Web Store Deployment (Step-by-Step)

Step 1: Package the Extension

1. **Set your admin password** (if you haven't already):
   - Edit `popup.js` line 32: `const ADMIN_PASSWORD = "YOUR_PASSWORD_HERE";`
   - Edit `options.js` line 13: `const ADMIN_PASSWORD = "YOUR_PASSWORD_HERE";`

2. **Create a `.zip` file** of the extension:
   ```
   bash
     # From your project directory
     zip -r mck-browser-manager.zip * -x "*.git*" "*.DS_Store" "node_modules/*"
   ```

Or manually:
- Select all files in `/mck-browser-manager/` folder
- Right-click → **Compress** (Mac) or **Send to → Compressed folder** (Windows)
- Name it `mck-browser-manager.zip`

⚠️ **Important:** Zip the **contents** of the folder, not the folder itself. When you unzip, you should see `manifest.json` at the root, not `mck-browser-manager/manifest.json`.

Step 2: Chrome Web Store Developer Registration

1. Go to Chrome Web Store Developer Dashboard (https://chrome.google.com/webstore/devconsole)
2. Sign in with your Google Workspace admin account (or a dedicated developer account)
3. **Pay the $5 one-time registration fee** (if not already registered)
4. Agree to the Developer Agreement

Step 3: Upload Your Extension

1. Click **"New Item"** in the Developer Dashboard
2. **Upload** your `mck-browser-manager.zip` file
3. Fill in the required listing information:

Store Listing Tab:
- **Name:** MCK Browser Manager
- **Summary:** Enterprise browser management for McKinney — one-click cache clearing, performance tuning, and privacy controls.
- **Description:**
```
MCK Browser Manager is an enterprise Chrome extension for McKinney employees,
providing centralized browser management tools.

Features:
• One-click cache clearing with visual feedback
• Auto-clear cache on Chrome startup (IT-managed)
• Performance controls: Memory Saver, Tab Hibernation, Preload Pages
• Privacy settings: Do Not Track, 3rd Party Cookie blocking, WebRTC protection
• Centrally managed by McKinney IT Help Desk

This extension is designed for internal use and managed via Google Admin Console.
```

- **Category:** Productivity
- **Language:** English (United States)

Upload screenshots (recommended):
- Take 2-3 screenshots of the extension popup (Cache tab, Privacy tab, etc.)
- Chrome Web Store requires at least 1 screenshot (1280x800 or 640x400)

Privacy Tab:
- **Single Purpose:** Browser management and performance optimization for enterprise users
- **Permission Justification:** Explain each permission:
- `browsingData`: Required to clear cache
- `storage`: Persist user settings
- `tabs`: Memory Saver and Tab Hibernation features
- `privacy`: Apply Do Not Track and WebRTC settings

- `contentSettings` : Block 3rd party cookies
- `alarms` : Schedule automated tab management

Visibility:
- Select **“Unlisted”** (users with the link can install, but it won’t appear in search)
- OR **“Private”** (only your Google Workspace domain can install) — requires domain verification

1. Click **“Submit for Review”**

Step 4: Wait for Approval

- **Review time:** Usually 1-3 business days (sometimes as fast as a few hours)
- You’ll receive an email when the extension is approved
- Once approved, note the **Extension ID** (looks like: `abcdefghijklmnopqrstuvwxyz123456`)

Step 5: Deploy via Google Admin Console

Once your extension is live in the Chrome Web Store:

1. **Sign in to Google Admin Console**
   Go to admin.google.com (https://admin.google.com)

2. **Navigate to Chrome Management**
   `Devices` → `Chrome` → `Apps & extensions`

3. **Add the Extension**
   - Click **“Add Chrome app or extension”** (yellow ➕ button)
   - Paste your **Extension ID** from the Chrome Web Store
   - Click **“Select”**

4. **Configure Installation Policy**
   - **Who:** Select the organizational unit (OU) to deploy to:
     - Select **“All users”** for company-wide deployment
     - Or choose specific departments/groups
   - **Installation Policy:** Choose **“Force install”**
     - This automatically installs the extension for all users in the selected OU
     - Users **cannot** uninstall it (true enterprise policy enforcement)
   - **Installation URL:** Auto-filled (Chrome Web Store URL)

1. **Pin to Toolbar (Optional but Recommended)**
   - Under “User Experience” settings:
     - Set **“Toolbar pin”** to **“Force pinned”**
     - This ensures the extension icon is always visible in the Chrome toolbar

2. **Configure Settings (Optional)**
   You can also pre-configure settings via Admin Console policies, but since your extension uses `chrome.storage.local` , users will have the default settings defined in `background.js` .

3. **Click “Save”**

Step 6: Rollout

- **Propagation time:** Changes can take up to 24 hours to apply to all devices

- **For faster testing:** Have a user:
  1. Close and reopen Chrome
  2. Or visit `chrome://policy` → Click **“Reload policies”**
- The extension will automatically install on all managed Chrome browsers within your selected OU
- Users will see “MCK Browser Manager” in their toolbar (if pinned) or extensions list

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Option B: Self-Hosted Deployment (Advanced)

If you don’t want to publish to Chrome Web Store:

Step 1: Package and Sign the Extension

1. **Create a `.crx` package** (Chrome Extension Package):
   ```
   bash
     # Chrome requires extensions to be signed
     # Use chrome://extensions/ → Pack extension
     # Or use command line (requires Chrome binary)
   ```
2. **Host the `.crx` and `update.xml` files** on an HTTPS server:
   - Upload `mck-browser-manager.crx` to your web server
   - Create an `update.xml` manifest (example below)

update.xml:
```xml

```

Step 2: Deploy via Admin Console

1. In Google Admin Console:
   `Devices` → `Chrome` → `Apps & extensions` → `Add Chrome app or extension`
2. **Choose “Add custom app”**
3. Enter:
   - **Extension ID:** (from your `.pem` file or generated during packing)
   - **Update URL:** `https://your-domain.com/update.xml`
4. Set **Installation Policy** to **“Force install”**
5. Click **“Save”**

⚠️ **Self-hosting drawbacks:**
- More complex initial setup
- You must manually update `update.xml` and `.crx` for every new version
- No automatic security/malware scanning by Google

Updating the Extension

If using Chrome Web Store:

1. Make your code changes locally
2. Increment the `version` in `manifest.json` (e.g., `1.0.0` → `1.0.1`)
3. Re-zip the extension
4. Upload to Chrome Web Store Developer Dashboard (same listing, click “Package” tab → “Upload new package”)
5. Submit for review
6. Once approved, updates **automatically roll out** to all installed instances (within 24-48 hours)

If self-hosting:

1. Update your code
2. Increment `version` in `manifest.json`
3. Re-create the `.crx` file
4. Update `update.xml` with the new version number and `.crx` URL
5. Upload both files to your server
6. Chrome checks for updates periodically (every few hours)

Testing the Deployment

Before full rollout:

1. **Create a test OU** in Google Admin:
   - `Admin Console` → `Directory` → `Organizational units` → **Create test OU**
   - Add 1-2 test users to this OU
   - Deploy the extension to this OU first with “Force install”

2. **Test user workflow:**
   - Sign in to Chrome as a test user
   - Verify the extension auto-installs
   - Check all features:
     - Cache clearing works
     - Admin password gate works (try toggling “Auto-clear on startup”)
     - Advanced Settings cog reveals tabs
     - Privacy/Performance toggles apply correctly

3. **Once validated:**
   - Expand deployment to **All users** or your main OUs

Troubleshooting

Extension not installing for users?

- Check the OU selection in Admin Console (ensure the user is in the target OU)

- Verify “Force install” is selected (not “Allow install”)
- Ask users to visit `chrome://policy` → **Reload policies** → check “ExtensionInstallForcelist”
- Check Chrome Management enrollment: `chrome://management/` should show “Your browser is managed by [your-domain]”

Users can uninstall the extension?

- Ensure you selected **“Force install”** (not “Allow install”) in Admin Console
- Force install prevents uninstallation by users

Settings not persisting?

- The extension uses `chrome.storage.local` which is tied to the Chrome profile
- If users are using temporary/guest profiles, settings won’t persist
- Ensure users are signed in to their Workspace account in Chrome

Admin password visible in source code?

- **This is intentional and expected for client-side extensions** — the password is a deterrent, not a security boundary
- For true enforcement, use Chrome Enterprise Policies (can lock settings via Admin Console)
- The current password gate is sufficient for preventing **accidental** changes by non-technical users

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Security Best Practices

1. **Restrict Admin Console access** to IT staff only
2. **Review extension updates** before deploying (if self-hosting)
3. **Monitor user feedback** via IT Help Desk link (`servicedesk.mckinney.com`)
4. **Use “Force install”** to prevent users from disabling/removing the extension
5. **Change the admin password** periodically (update in both `popup.js` and `options.js` , re-deploy)

Support & Maintenance

- **IT Help Desk:** https://servicedesk.mckinney.com/support/home (https://servicedesk.mckinney.com/support/home)
- **Extension version:** Displayed in footer of popup and options page
- **Chrome version compatibility:** Manifest V3 requires Chrome 88+ (2021), all modern Chrome browsers supported

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Quick Reference

Task	Google Admin Console Path
Deploy extension	`Devices` → `Chrome` → `Apps & extensions` → `Add app`
Pin to toolbar	Extension settings → `Toolbar pin` → `Force pinned`
View installed extensions	`Devices` → `Chrome` → `Apps & extensions`
Check user devices	`Devices` → `Chrome` → `Devices`
Force policy update	User visits `chrome://policy` → `Reload policies`

Additional Resources

- Chrome Enterprise Policy List (https://chromeenterprise.google/policies/)
- Chrome Web Store Developer Documentation (https://developer.chrome.com/docs/webstore/)
- Google Workspace Admin Help: Manage Chrome Apps (https://support.google.com/chrome/a/answer/2657289)

Questions? Contact your IT Help Desk or the extension developer.